Tony Nguyen

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Education

M.S. in Artificial Intelligence, *Machine Learning* — *Expected May 2028*
Northeastern University — Seattle, WA

B.A. in Computer Science, *Data Science & Economics* — *May 2026*
B.B.A. in Business Administration, *Finance* — Spokane, WA
Gonzaga University
President's Honor List, *Summa Cum Laude*, GPA: 3.95

Research Interests

Knowledge representation and evaluation in LLMs. Skill integration for AI-assisted threat hunting. Retrieval-augmented generation, agentic systems, and resource-constrained deployment for reliable AI.

Research & Industry Experience

AI / Machine Learning Research Assistant (Advisor: Dr. Jay Yang) — January 2025 - May 2026
Institute for Informatics & Applied Technology - Gonzaga University — *Spokane, WA*

- Designed and built the **Wazuh MCP Server**, an open-source Python server (**30+ endpoints**) enabling LLMs to query, investigate, and analyze security events from a Wazuh SIEM platform through structured tool calls.
- Designed a controlled **benchmarking study** comparing **skill integration** across two frontier models (Claude Sonnet and Opus) against a **ground-truth attack chain**, showing that structured domain knowledge closes the performance gap between smaller and larger models, and extended the study to **Gemma 4 (31B)** on local GPU infrastructure to test generalization to open-source models.
- Benchmarked **three LLMs** on **more than 60 CVE reports** using **a three-stage RAG pipeline** with specialized **agents** for retrieval, analysis, and remediation recommendation, focusing on insightfulness and verifiability.
- Built and benchmarked ZagAI, an RAG study assistant for Gonzaga faculty and students, comparing retrieval and chunking strategies and instrumenting the pipeline with observability to inspect retrieval and agent behavior.
- Interviewed 5 faculty and 7 students to define requirements, and iterated the design on their feedback so the system fit how each group teaches and studies.
- Developed and delivered a workshop on **RAG** systems and **LLM evaluation** to over **30 participants**, covering classical metrics, agent-based evaluation, and responsible AI practices.

Cofounder, Business Lead — August 2026 – Present
kapsl — *Seattle, WA*

- Co-founded a **Rust-based** AI inference runtime that governs **GPU memory** dynamically across concurrent models, positioned against static-partitioning approaches (Triton, vLLM) for shared-GPU inference workloads.
- Designed a **benchmarking methodology** for the runtime with repeated runs, statistical validation, and hashed provenance records for every result. Stress-tested memory admission control under concurrent load and over-requests, with no crashes and no measurable impact on co-resident models.

Data Engineer Intern — June 2025 - September 2025
KPIM Vietnam — *Hanoi, Vietnam*

- Designed and maintained **10+ data pipelines** using **SQL, Python, and Microsoft Fabric** for **2 Big Data projects**, enforcing complex schema validation, data quality checks, and optimizations to make querying **50% faster** than the client's internal system.
- Built reusable tooling and documentation to streamline troubleshooting, standardize workflows, and support stakeholders with reliable, self-serve financial and operational data of the client's **6 hotel properties**.

Data Engineer - Innovation Intern — June 2024 - August 2024
KONE Elevators & Escalators — *Singapore*

- Built **a centralized Customer 360** by integrating **20 years** of data from **5 countries**; aligned models to enterprise architecture and recommended schema optimizations to improve consistency and join performance.
- Developed a **Tableau** dashboard that fed from **SAP, Salesforce, and Excel**; created data models and calculations to monitor customer behavior, service KPIs, and revenue in real time.

· Engineered and operated **ETL** pipelines on **AWS S3** for ingestion, storage, and validation; automated integrity checks and partitioning to **reduce data retrieval latency by 40%** and improve reliability.

Data Science & Machine Learning Intern June 2023 - August 2023
TecAlliance Vietnam *Ho Chi Minh City, Vietnam*

· Engineered Python algorithms to parse data from **100+ Wiring Diagram SVGs**, creating structured datasets for computer vision models (**YOLO**) and advanced ML pipelines.
· Deployed **a Neo4j graph database** to store complex relationships extracted from automotive schematics, increasing **query speed by 80%** compared to the client's current database and providing deeper insights.
· Led weekly team meetings of **15 people** across time zones to unify data engineering efforts.

Conference Poster Presentations

· Value & Responsibility in AI Technologies Conference, 2026. Spokane, WA.
· University of Washington Undergraduate Research Symposium, 2026. Seattle, WA.
· Value & Responsibility in AI Technologies Conference, 2025. Spokane, WA.
· Spokane Intercollegiate Research Conference, 2025. Spokane, WA.

Projects

Minecraft AI Companion – Senior Capstone Project August 2025 - May 2026

· Built an open-source Minecraft mod using a **two-model architecture** (local LLM for real-time dialogue, cloud model for retrieval-augmented queries), designing the context pipeline, tool-calling interface, and content guardrails for safe in-game interactions.

Mortality Risk Simulator January 2025 - May 2025

· Led a **neural-network** mortality risk pipeline (**PyTorch**) over demographic, vitals, and history features.

Teaching & Mentoring Experience

Teaching Assistant August 2024 - May 2026
Department of Computer Science - Gonzaga University *Spokane, WA*

· Supported **150+** undergraduates across **3 data science courses**, leading weekly office hours to reinforce students' Python, pandas, and scikit-learn skills and core algorithms and data workflows.
· Graded **10 programming assignments each term** using shared rubrics, providing targeted feedback on correctness, code structure, and documentation to improve code quality and conceptual understanding.

Academic Tutor & Mentor October 2022 - May 2026
Gonzaga Global - Gonzaga University *Spokane, WA*

· Provided tutoring and lab support for **7 computer science and quantitative courses** for **dozens of students each term**, contributing to **15%+ average grade improvement** for tutees.
· Mentored international students on U.S. coursework, study strategies, and academic integrity.

Professional & Leadership Experience

Academic Coordinator September 2026 - Present
Global Experience Office - Northeastern University *Seattle, WA*

· Leading requirements and engineering to replace a 20-year-old NUin class registration system, mapping workflows and undocumented workarounds across 8 stakeholder groups to define system requirements and integrations.

Administrative Assistant March 2022 - May 2026
Center for Global Engagement - Gonzaga University *Spokane, WA*

· Mentored **50+** incoming international students through cultural and academic transition to US college; designed automated workflows reducing administrative burden by **30%**, enabling more direct student support.

Technical Proficiencies

Languages & ML	Python, SQL, C++, Java; PyTorch, scikit-learn, Hugging Face Transformers, pandas, NumPy
LLM & Systems	RAG, agentic systems, LLM evaluation, MCP, LangChain, CrewAI; FastAPI, Flask, Docker, Git, Linux, AWS